Classical Probability

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Intuitive probability

- Probability – the chance that an uncertain event will occur (always between 0 and 1)

Symbols:

P(event A) = "the probability that event A will occur"

P(red card) = "the probability of a red card"

Sample space and Events

- Ω : Sample Space, result of an experiment
  - If you toss a coin twice $\Omega = \{\mathrm{HH},\mathrm{HT},\mathrm{TH},\mathrm{TT}\}$
- Event: a subset of Ω
  - First toss is head = {HH,HT}
- S: event space, a set of events:
  - Closed under finite union and complements
    - Entails other binary operation: union, diff, etc.
  - Contains the empty event and Ω

Probability Measure

- Defined over (Ω,S) s.t.
  - $P(\alpha) >= 0$ for all α in S
  - $P(\Omega) = 1$
  - If $\alpha_1,\ldots,\alpha_N$ are disjoint events, then

    $P(U\alpha_i) = p(\alpha_1) +\ldots+ p(\alpha_N)+\ldots$

<u>Remark:</u>

- We can deduce other properties from the above ones.

Important properties of probability

- $P(\emptyset)=0$
- Finite additivity
- Monotonicity
- Addition formula

Assessing Probability

1. Theoretical/Classical probability—based on theory (*a priori* understanding of a phenomena)

   e.g.: theoretical probability of rolling a 2 on a standard die is 1/6

   theoretical probability of choosing an ace from a standard deck is 4/52

   theoretical probability of getting heads on a regular coin is 1/2

2. Empirical probability—based on empirical data

   e.g.: you toss an irregular die (probabilities unknown) 100 times and find that you get a 2 twenty-five times; empirical probability of rolling a 2 is 1/4

   empirical probability of an Earthquake in Bay Area by 2032 is .62 (based on historical data)

   empirical probability of a lifetime smoker developing lung cancer is 15 percent (based on empirical data)

Computing theoretical probabilities

If <u>outcomes are equally likely to occur</u>…

$$P(A) = \frac{\text{\# of ways A can occur}}{\text{total \# of outcomes}}$$

<u>Note:</u> This is called "counting methods" in which we have to <u>**count**</u> the number of ways A can occur and the number of total possible outcomes.

Counting methods: Example 1

Example 1: You draw one card from a deck of cards. What's the probability that you draw an ace?

$$P(\text{draw an ace}) = \frac{\#\text{ of aces in the deck}}{\#\text{ of cards in the deck}} = \frac{4}{52} = .0769$$

Counting methods: Example 2

Example 2. What's the probability that you draw 2 aces when you draw two cards from the deck?

$$P(\text{draw ace AND ace}) = \frac{4}{52} \text{ x } \frac{3}{51}$$

Counting methods: Example 2

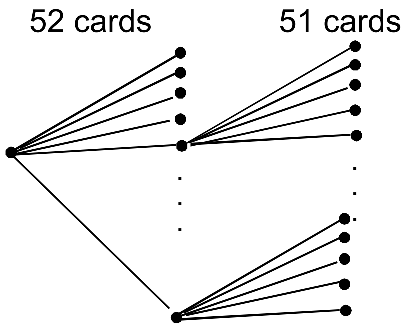
<u>Two counting method ways to calculate this:</u>

1. Consider order:

$$P(\text{draw 2 aces}) = \frac{\text{\# of ways you can draw ace, ace}}{\text{\# of different 2-card sequences you could draw}}$$

Numerator: A♣A♦, A♣A♥, A♣A♠, A♦A♥, A♦A♠, A♦A♣, A♥A♦, A♥A♣, A♥A♠, A♠A♣, A♠A♦, or A♠A♥ = 12

Denominator = 52x51 = 2652



$$\therefore P(\text{draw 2 aces}) = \frac{12}{52x51}$$

Counting methods: Example 2

2. Ignore order:

$$P(\text{draw 2 aces}) = \frac{\text{\# of pairs of aces}}{\text{\# of different two - card hands you could draw}}$$

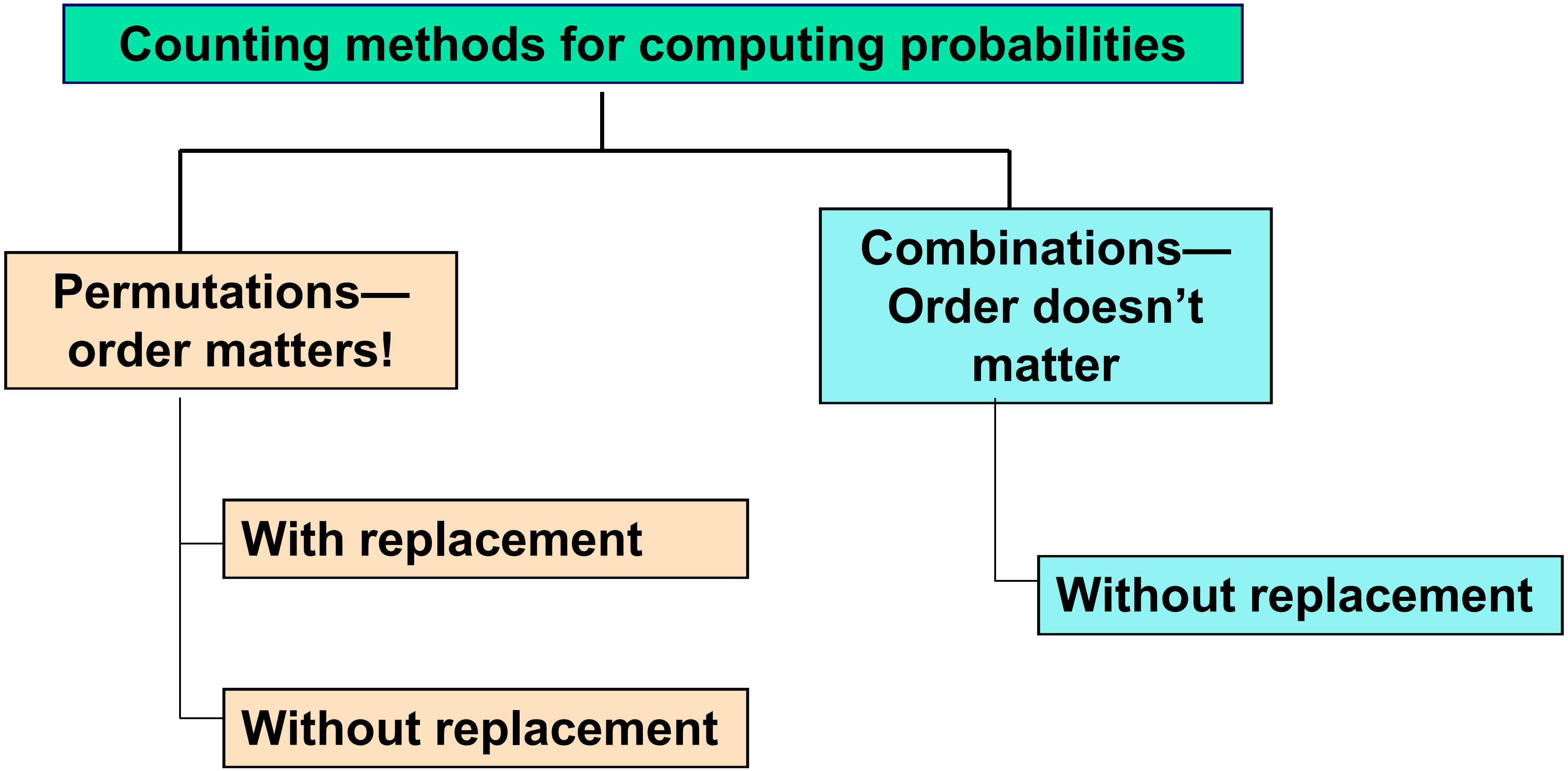
Numerator: A♣A♦, A♣A♥, A♣A♠, A♦A♥, A♦A♠, A♥A♠ = 6

$$\text{Denominator} = \frac{52x51}{2} = 1326$$

Divide out order!

$$\therefore P(\text{draw 2 aces}) = \frac{6}{\frac{52x51}{2}}$$

Summary of Counting Methods



Permutations—Order matters!

A permutation is an ordered arrangement of objects.

<u>With replacement</u>=once an event occurs, it can occur again (after you roll a 6, you can roll a 6 again on the same die).

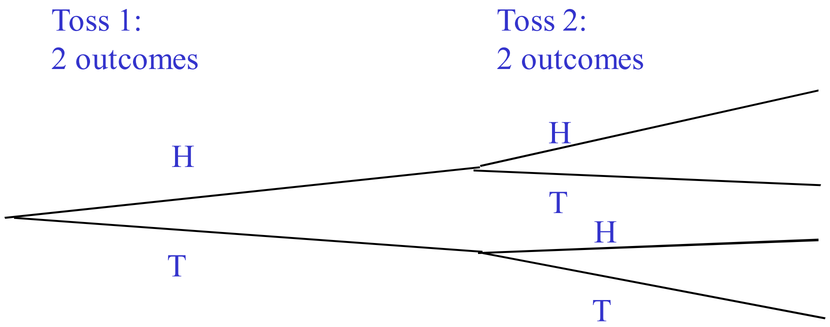
<u>Without replacement</u>=an event cannot repeat (after you draw an ace of spades out of a deck, there is 0 probability of getting it again).

Permutations—with replacement

With Replacement – Think coin tosses, dice, and DNA.

"memoryless" – After you get heads, you have an equally likely chance of getting a heads on the next toss (unlike in cards example, where you can't draw the same card twice from a single deck).

What's the probability of getting two heads in a row ("HH") when tossing a coin?



2^2 total possible outcomes: {HH, HT, TH, TT}

$$P(HH) = \frac{\text{1 way to get HH}}{2^2 \text{ possible outcomes}}$$

Formally, “order matters” and “with replacement” → use powers →

$$\boxed{(\text{\# possible outcomes per event})^{\text{the \# of events}} = n^r}$$

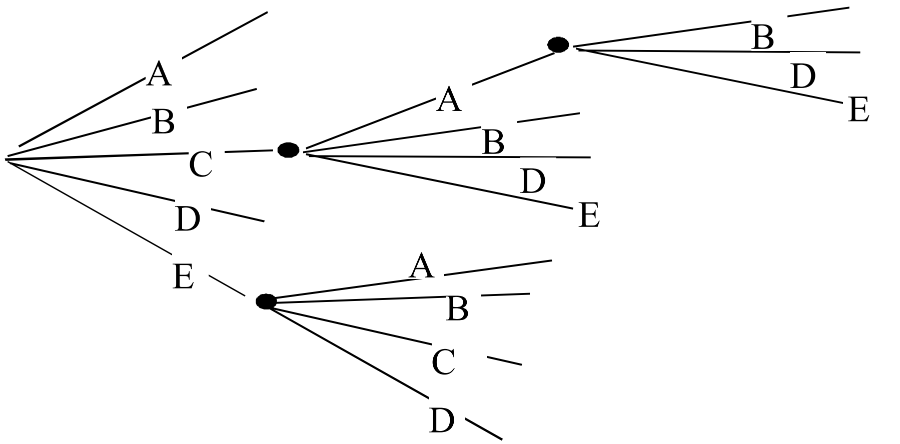
Permutation—without replacement

What if you had to arrange 5 people in only 3 chairs (meaning 2 are out)?

<u>Seat One:</u>
5 possible

<u>Seat Two:</u>
Only 4 possible

<u>Seat Three:</u>
only 3 possible



$$\boxed{5x4x3 =}$$

$$\boxed{\frac{5x4x3x2x1}{2x1} = \frac{5!}{2!} =}$$

$$\boxed{\frac{5!}{(5-3)!}}$$

Formally, "order matters" and "without replacement"→ use factorials→

$$\frac{(n\ \text{people or cards})!}{(n\ \text{people or cards} - r\ \text{chairs or draws})!} = \frac{n!}{(n-r)!}$$

$$\text{or}\quad n(n-1)(n-2)\ldots(n-r+1)$$

2. Combinations—Order doesn't matter

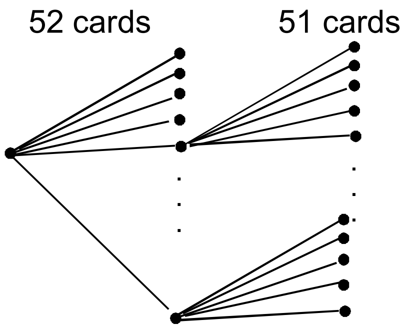
Introduction to combination function, or "choosing"

Written as: C_n^r or $\binom{n}{r}$

Spoken: "*n* choose *r*"

Combinations

How many two-card hands can I draw from a deck when order does <u>not</u> matter (e.g., ace of spades followed by ten of clubs is <u>the same</u> as ten of clubs followed by ace of spades)



$$\boxed{\frac{52x51}{2} = \frac{52!}{(52-2)!2}}$$

Formally, "order doesn't matter" and "without replacement"→ use choosing→

$$\boxed{\binom{n}{r} = \frac{n!}{(n-r)!\,r!}}$$

Examples—Combinations

A lottery works by picking 6 numbers from 1 to 49. How many combinations of 6 numbers could you choose?

$$\binom{49}{6} = \frac{49!}{43!6!} = 13{,}983{,}816$$

Which of course means that your probability of winning is 1/13,983,816!

Examples

How many ways can you get 3 heads in 5 coin tosses?

$$\binom{5}{3} = \frac{5!}{3!2!} = 10$$

Counting methods for computing probabilities

Permutations—order matters!

With replacement: n^r

Without replacement: $n(n-1)(n-2)\ldots(n-r+1)=$

$$\frac{n!}{(n-r)!}$$

Combinations—Order doesn't matter

Without replacement:

$$\binom{n}{r}=\frac{n!}{(n-r)!r!}$$

Practice problem:

- A classic problem: "The Birthday Problem." What's the probability that two people in a class of 25 have the same birthday? (disregard leap years)

  What would you guess is the probability?

Birthday Problem Answer

1. A classic problem: "The Birthday Problem." What's the probability that two people in a class of 25 have the same birthday? (disregard leap years)

***Trick!* 1- P(none) = P(at least one)

Use complement to calculate answer. It's easier to calculate 1- P(no matches) = the probability that at least one pair of people have the same birthday.

What's the probability of no matches?

Denominator: how many sets of 25 birthdays are there?

--with replacement (order matters)

365^{25}

Numerator: how many different ways can you distribute 365 birthdays to 25 people without replacement?

--order matters, without replacement:

[365!/(365-25)!]= [365 x 364 x 363 x 364 x ….. (365-24)]

$\therefore$ P(no matches) = [365 x 364 x 363 x 364 x ….. (365-24)] / 365^{25}

Birthday Problem Answer

$$P(A) = 1 - P(A')$$

$$\# A' = P_{365}^{n}$$

$$P(A') = \frac{\# A'}{\# \Omega}$$

$$\# \Omega = 365^{n}$$

Birthday Problem → Birthday Paradox

- How many people is required to reach a probability of 50% that at least two individuals in the group have the same birthday?

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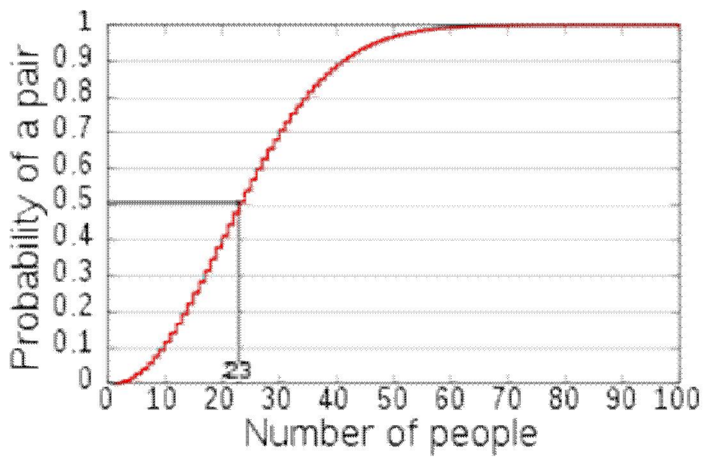
$$P(A) = 1 - \frac{365!}{365^n (365-n)!} = 0.5$$



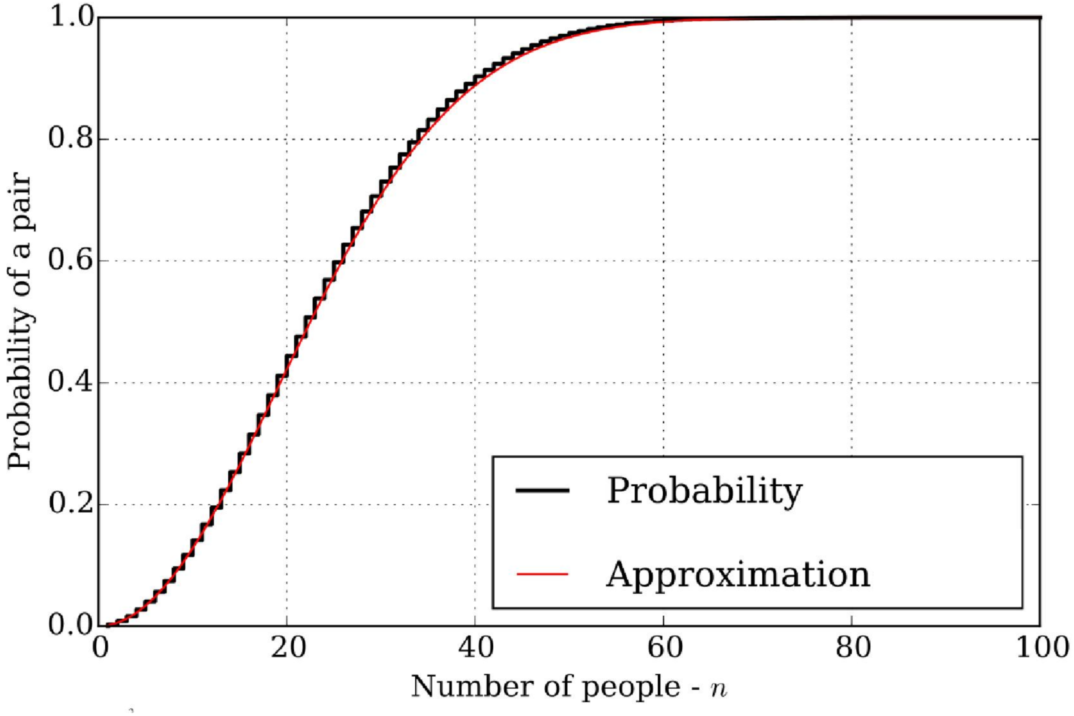
Approximation

$$\frac{365!}{365^n(365-n)!} = (1-\frac{1}{365}) \times (1-\frac{2}{365}) \times \cdots \times (1-\frac{n-1}{365})$$

$$1\text{-}x \approx e^{-x} \qquad (|x| \ll 1)$$

$$(1-\frac{1}{365}) \times (1-\frac{2}{365}) \times \cdots \times (1-\frac{n-1}{365}) \approx e^{-1/365} \times \ldots \times e^{-(n-1)/365} = e^{-n(n-1)/730}$$

The approximation is pretty good!



Generalization as a collision problem

- Given *n* random integers drawn from a discrete uniform distribution with range [1,d], what is the probability *p(n; d)* that at least two numbers are the same? (d = 365 gives the usual birthday problem)

$$p(n;d) = \begin{cases} 1 - \prod_{k=1}^{n-1}\left(1 - \frac{k}{d}\right) & n \le d \\ 1 & n > d \end{cases}$$

$$\approx 1 - e^{-\frac{n(n-1)}{2d}}$$

Application - cryptographic attack（密码攻击）

- Suppose that a function *f(x)* yields any of *H* different outputs with equal probability. The goal of the attack is to find two different inputs *x1* and *x2* such that *f(x1)=f(x2)*. Such a pair *x1*, *x2* is called **a collision**.

- From a set of *H* values (*H* is sufficiently large) we choose *n* values uniformly at random with repetition. Let $p(n; H)$ be the probability that at least one value is chosen more than once, then

$$p(n;H) \approx e^{-n(n-1)/2H} \approx e^{-n^2/2H}$$

Application - cryptographic attack（密码攻击）

- The smallest number of values we have to choose such that the probability for finding a collision is at least *p*

Bits	Possible outputs (H)	Desired probability of random collision (2 s.f.) (p)									
		10^{-18}	10^{-15}	10^{-12}	10^{-9}	10^{-6}	0.1%	1%	25%	50%	75%
16	2^{16} (~6.5×10^{4})	<2	<2	<2	<2	<2	11	36	190	300	430
32	2^{32} (~4.3×10^{9})	<2	<2	<2	3	93	2900	9300	50,000	77,000	110,000
64	2^{64} (~1.8×10^{19})	6	190	6100	190,000	6,100,000	1.9×10^{8}	6.1×10^{8}	3.3×10^{9}	5.1×10^{9}	7.2×10^{9}
128	2^{128} (~3.4×10^{38})	2.6×10^{10}	8.2×10^{11}	2.6×10^{13}	8.2×10^{14}	2.6×10^{16}	8.3×10^{17}	2.6×10^{18}	1.4×10^{19}	2.2×10^{19}	3.1×10^{19}
256	2^{256} (~1.2×10^{77})	4.8×10^{29}	1.5×10^{31}	4.8×10^{32}	1.5×10^{34}	4.8×10^{35}	1.5×10^{37}	4.8×10^{37}	2.6×10^{38}	4.0×10^{38}	5.7×10^{38}
384	2^{384} (~3.9×10^{115})	8.9×10^{48}	2.8×10^{50}	8.9×10^{51}	2.8×10^{53}	8.9×10^{54}	2.8×10^{56}	8.9×10^{56}	4.8×10^{57}	7.4×10^{57}	1.0×10^{58}
512	2^{512} (~1.3×10^{154})	1.6×10^{68}	5.2×10^{69}	1.6×10^{71}	5.2×10^{72}	1.6×10^{74}	5.2×10^{75}	1.6×10^{76}	8.8×10^{76}	1.4×10^{77}	1.9×10^{77}

Further questions

- If the outputs of the function are distributed unevenly, is a collision found faster or slower?
- What is the expected number of values we have to choose before finding the first collision?
- Optional: same birthday → similar birthday